

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0187

Wurno 100 kV
February 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0187. The Interconnection Customer proposes a 291.6 MW natural gas combined cycle facility interconnecting at the Wurno 100 kV substation in the Illinois area. The request was received on December 20, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 113 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,903,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Wurno – Pgg - Prince G 100 KV: 200.1% loading (118 MVA) following the outage of the Wurno – Bellwood 100 KV; DFAX = 100.0%.

For the loss of Wurno – Pgg - Prince G 100 KV L, the Wurno – Bellwood 100 KV loads to 184.6% of its 127 MVA rating. Project DFAX: 100.0%.

Bellwood – Antigo 100 KV: 144.8% loading (138 MVA) following the outage of the Wurno – Pgg - Prince G 100 KV; DFAX = 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wurno – Pgg - Prince G 100 KV	Rebuild line with higher-capacity conductor	\$1,807,000	100% (\$1,807,000)
Wurno – Bellwood 100 KV	Rebuild line with higher-capacity conductor	\$2,667,000	100% (\$2,667,000)
Bellwood – Antigo 100 KV	Rebuild line with higher-capacity conductor	\$3,429,000	100% (\$3,429,000)
	Total allocated to project		\$7,903,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.